

Unified Model of Current-Limiting Grid-Forming Inverters for Large-Signal Analysis

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Abstract—Grid-forming (GFM) inverters can hardly withstand any overloading. As such, GFM inverters need a current limiter in their control system to avoid hardware damage during overloading or grid disturbances. When the current limiter is engaged for inverter protection, however, it dynamically changes the inverter output impedance, which can significantly affect the system behavior against large disturbances. For example, when the current limiter curtails the output power, the primary controller of the GFM inverter—which generates the voltage and angle reference—is prone to losing synchronization with the grid. This can cause large-signal instability in the network. This paper presents a unified GFM current-limiter model to gain a deeper understanding of the impact of the GFM inverter current limiting on large-signal instability and other system behaviors. The unified model captures several well-known current limiters for GFM inverters, and it reveals their large-signal equivalence characterized by the limiter angle. We use the unified model to present in-depth insights into the driving forces that cause large-signal instability in GFM inverter-driven grids. Further, this paper illustrates that the unified model can be used as a tool for assessing the system-level impacts of various GFM current-limiter designs during grid disturbances. Using numerical, analytical, electromagnetic transient models, and a hardware setup, the validity of the unified model is evaluated.

Index Terms—Grid-forming inverter current limiting, large-signal stability, equivalent-circuit model, power-angle curve, virtual impedance, power system transient stability.

I. INTRODUCTION

GRID-FORMING (GFM) inverter research and development has gained significant traction in recent years. Unlike their contemporary grid-following (GFL) counterpart, GFM inverters create their own voltage and frequency reference, enabling grids to operate at high levels of inverter-based resources (IBRs). For these reasons, adopting GFM capabilities in grid-connected inverters has been urgently called for by system operators, research institutions, and companies around the world [1], [2],

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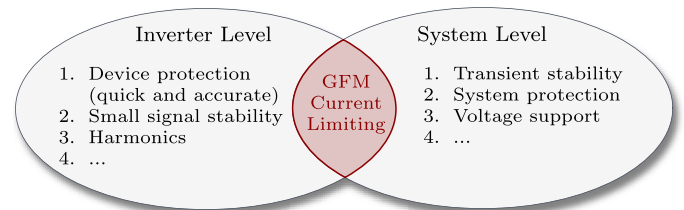


Fig. 1. Inverter-level and system-level ramifications of GFM inverter current limiting [13].

[3], [4], [5], [6]. However, the transition from synchronous-generator (SG) dominated grids toward grids with (GFM) IBRs proves challenging, and many research questions are yet to be answered [7], [8]. Especially determining how GFM inverters should behave during grid disturbances (such as (un)balanced faults, frequency or phase jumps, or overloading), their system-level influence (small- and large-signal stability and power system protection), and how to design the controls accordingly are ongoing research topics [9], [10], [11]. At the center of these research questions lies the inability of inverters to inject high fault currents. Power-processing components of power electronics inverters, such as semiconductor switches, have limited overloading capabilities that barely exceed the nominal values—typically ranging from 1.05 pu–1.20 pu—because of their low thermal inertia and the design practice for cost optimization [12]. To that end, a current limiter is an essential block in the inverter control system which strictly limits the output current to not cross hardware limitations that can cause irreversible inverter damage. Although the existential purpose of the limiter is solely to prevent the inverter output currents from exceeding the acceptable value, the limiter design profoundly influences the entire dynamic behavior of the inverter during current-limited conditions. Namely, because the current limiter dynamically alters the output impedance of the inverter, network-wide attributes for a power system, such as transient stability, power system protection, and voltage support, can be compromised. The device-level and system-level impacts of the GFM inverter current limiter are visually represented in Fig. 1, and it illustrates the need to investigate the GFM current limiting to understand system resilience during large disturbances in an IBR-dominated grid [13].

Because the limiter curtails the output current—and hence reshapes the active and reactive power output—it is the main contributor to GFM inverter instability during grid

disturbances [14], [15]. This issue has been discussed in the literature: In [16], [17], [18], the pernicious effects of GFM current limiting on transient stability are illustrated through simulations and hardware experiments. These works assume idealized scenarios without filter or line resistance or assume a current limiter where the current-phaser angle is fixed. In [19], the authors present an equivalent circuit model to compile the power-angle characteristic for a GFM inverter with a circular current-reference limiter and validate the theoretical results through hardware tests. In [20], a similar study is presented for virtual impedance (VI)-based current limiters. The authors illustrate that inductive VI limiters inherently induce better stability margins compared to resistive VI limiters, which we further acknowledge in this work. In an attempt to guarantee GFM inverter transient stability, despite the stringent current limits, other works have proposed additional control blocks to the primary controller, which—directly or indirectly—modify the perceived output power of the inverter and with it, the power-angle dynamics [21], [22], [23], [24], [25], [26]. A comparative overview of these methods is given in [27].

To better map and understand the influence of GFM current limiting on large-signal stability, this paper presents a unified model framework for GFM current-limiting methodologies. In that way, this work has a similar approach to [28], where the authors present a unified equivalent-circuit model that captures circular current limiting and virtual impedance limiting. In addition to the work in [28], this paper presents deeper understandings into the driving mechanisms behind large-signal instability with GFM inverters, based on novel theoretical derivations, and provides an in-depth sensitivity analysis of various attributes that induce instability. Initiated from the unified current-limiting framework, we aim to provide answers to the following questions: How can we intuitively understand the power-angle characteristics of GFM inverters under various types of current limiting? How do various current-limiting methods affect GFM inverter transient stability, both on a device level and system level? How can we derive an analytical model that captures various current-limiting methods to understand the power-angle dynamics of GFM inverters and derive critical clearing angles and times? With these questions in mind, this paper presents the following contributions:

- C1 Proposing a unified GFM inverter current limiter model that captures the large-signal behavior of GFM inverters with inner control loop-based current limiters, including the current-reference saturation limiter (also known as the circular current limiter or magnitude limiter) with anti-windup feedback, and the VI limiter. This generalized framework can be used to perform theoretical analyses of grid-connected GFM inverters under various current-limiting scenarios, such as large-signal stability analysis.
- C2 Deriving a numerical and analytical model for the power-angle characteristics of GFM inverters under current limitations. This provides a deeper understanding of the significant impact of current limiting on the GFM inverter stability during grid disturbances and the influence of control parameters and grid conditions.

- C3 Presenting full-order electromagnetic transient (EMT) simulations—both of a single-inverter and a realistic multi-inverter network simulation—along with a hardware setup, to support the theoretical findings on the large-signal stability of GFM inverters under current-limited conditions.

The rest of this paper is structured as follows. In Section II, we begin by introducing pertinent notations used throughout the paper. In Section III, we introduce the generic GFM inverter control architecture which will serve as a baseline throughout this work, and propose the concept of the unified current limiter framework. Following, in Section IV, we set up a numerical and analytical model to characterize the current-limited GFM inverter while encompassing all current limiter variations within the unified framework. Based on the analytical model, we provide an in-depth analysis of GFM inverter power-angle characteristics under current limiting and examine limiter dynamics under various grid conditions. In Section V, we dive deeper into the mechanism driving GFM inverter large-signal instability under various current-limiting strategies. We visualize and validate the numerical and analytical results through full-order EMT simulations—both in a single-inverter and multi-inverter network simulation—and a hardware setup of various grid disturbances. Finally, Section VI concludes the paper.

II. NOTATION AND PRELIMINARIES

In this work, complex phasors are denoted by $X_x \in \mathbb{C}$, with $|X_x| \in \mathbb{R}_{\geq 0}$ denoting the amplitude, and the angle denoted by $\angle X_x$ or $\varphi_x \in \mathbb{R}$. The complex conjugate operator is denoted by $\bar{X}$. The complex imaginary unit is denoted by j . We define a vector in the direct-quadrature plane as $x_{dq} \in \mathbb{R}^{2 \times 1} = [x_d, x_q]^\top$ and signals in the natural reference frame as $x_{abc} \in \mathbb{R}^{3 \times 1} = [x_a, x_b, x_c]^\top$. On occasion, we use the same notation to denote complex phasors as a vector, such that $X_{dq} \in \mathbb{R}^{2 \times 1} = [X_d, X_q]^\top$, with X_d and X_q representing the real and imaginary parts of the complex phasor, $X \in \mathbb{C}$. Scalars are denoted by $x \in \mathbb{R}$ and can be fixed or time/parameter varying. Rotation of a vector counterclockwise by angle θ in the Euclidean space is captured by:

$$T(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}.$$

We define the unit vectors as $e_1 = [1, 0]^\top$ and $e_2 = [0, 1]^\top$. Notations not captured in this section are explicitly denoted in the text.

III. UNIFIED GFM-INVERTER CURRENT-LIMITING FRAMEWORK

In this section, we introduce the GFM inverter control structure that will serve as the baseline for all our analyses throughout this work, and we introduce a unified current-limiting framework that captures two well-established current-limiting strategies in one model.

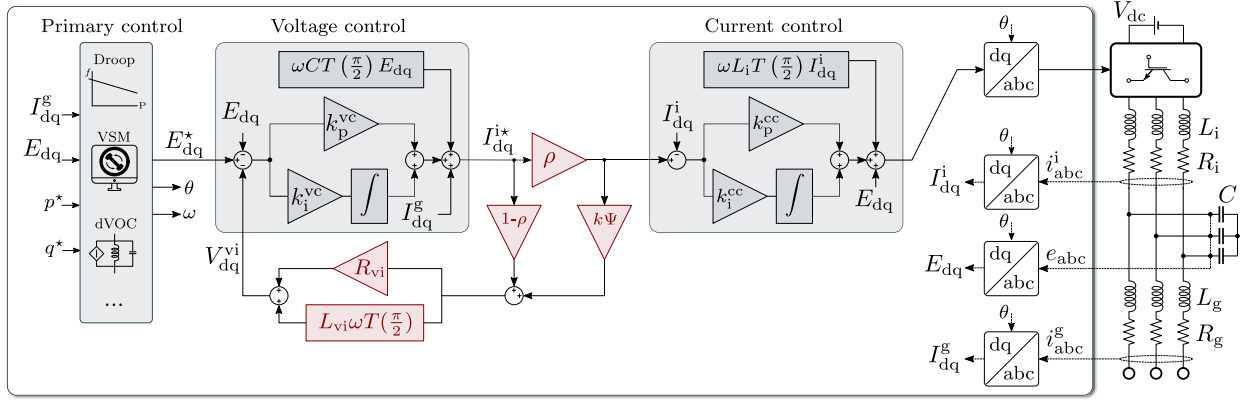


Fig. 2. GFM inverter control structure with the unified current limiter model.

A. GFM Inverter Control Design

Fig. 2 shows a prototypical GFM inverter control structure, built in the synchronous reference frame (dq frame), comprising inner current and outer voltage controllers, and a primary controller that governs the internal reference angle, frequency, and voltage magnitude. In this study, we consider the droop primary controller that is characterized by:

$$\frac{d\theta}{dt} = \omega_0 + m_p(p^* - \tilde{p}), \quad (1)$$

$$e^* = e_0 + m_q(q^* - \tilde{q}), \quad (2)$$

where θ denotes the instantaneous internal GFM inverter reference angle; ω_0 denotes the nominal grid angular frequency; m_p and m_q denote the droop gains; $\tilde{p}$ and $\tilde{q}$ denote the low-pass filtered active and reactive output power, respectively, with p^* and q^* denoting the active and reactive power setpoints, respectively; e^* and e_0 respectively denote the reference and nominal terminal voltage magnitude. In the control system, the sensed LCL filter currents and voltages are transformed using the Park transformation, denoted by I_{dq}^i , E_{dq} , and I_{dq}^g . The LCL filter parameters are denoted by L_i , R_i , C , L_g , and R_g . The inner-current and outer-voltage controllers employ PI controllers for which k_p^{cc} , k_p^{vc} , k_i^{cc} , and k_i^{vc} denote their proportional and integral gains, respectively. In the cross-coupling terms between the d- and q-axis, the matrix $T(\theta)$ denotes the counterclock rotation matrix, as defined in Section II. The unified current limiter representation is depicted in red, which we introduce next.

B. Unified GFM-Inverter Current-Limiting Framework

Hereafter, we introduce the unified GFM-inverter current-limiter model. Deriving such model has several merits. It allows for building one theoretical model that captures two common current-limiter variations without the need for mode switches. This simplifies implementation in particular for numerical computations. Leveraging the unified model, we can derive both a numerical and analytical expression for the power-angle relation of a GFM inverter, which further deepens the understanding of the large-signal behavior and stability of GFM inverters and the attributes influencing instability.

The unified current limiter structure is highlighted in red in Fig. 2. The model captures current-reference saturation limiting with an integrator anti-windup in the voltage controller (SatLim) [29], sometimes also referred to as a circular current limiter [19], and VI-based limiting with a threshold function (VI-Lim) [9]. In the unified current limiter, ρ denotes the current-reference saturation gain, which we define as:

$$\rho = \min \left(1, \frac{i_{\max}}{|I_{dq}^*|} \right). \quad (3)$$

The gain, ρ , ensures that the reference current feeding into the current controller is saturated to $i_{\max}$. The function, Ψ , denotes a threshold function to activate (or deactivate) the VI-Lim. We define Ψ as used in [9]:

$$\Psi = \max \left(0, \frac{\rho |I_{dq}^*| - \sigma}{i_{\max} - i_{th}} \right), \quad (4)$$

where σ denotes a parameter to enable or disable the VI-Lim, and i_{th} denotes the threshold current upon which the VI-Lim starts curtailing the current. The VI impedance in the unified model—denoted by Z_{vi} with angle φ_{vi} —comprises a resistive part, R_{vi} , and an inductive part, L_{vi} , and is cast in the dq frame as illustrated in Fig. 2. We tune the VI magnitude according to classical anti-windup tuning techniques, such that $|Z_{vi}| = (k_p^{vc})^{-1}$. The VI angle is tuned depending on the type of limiter (see further in this section). Last, we set the gain, $k = \epsilon_v k_p^{vc} i_{\max}^{-1}$, where ϵ_v denotes the maximum possible voltage error, $|E_{dq}^* - E_{dq}|_{\max}$, that can feed into the outer-voltage controller, which occurs when the primary controller is 180° phase shifted relative to a healthy grid voltage at the inverter terminals, in which case $\epsilon_v = 2$ pu. By tuning the gain, k , in this manner, we ensure that, when current limiting with the VI-Lim, the output current cannot surpass $i_{\max}$ even during the absolute worst-case scenario (except during excessive voltage swells, which requires more conservative (greater k) design) [9]. As illustrated in Fig. 3, we can obtain either the SatLim or VI-Lim by tuning the parameters σ and φ_{vi} in the unified current limiter model. If we set $\sigma \geq i_{\max}$ we retrieve the SatLim, as shown in Fig. 3(a). Namely, for $\sigma \geq i_{\max}$, the function $\Psi = 0$

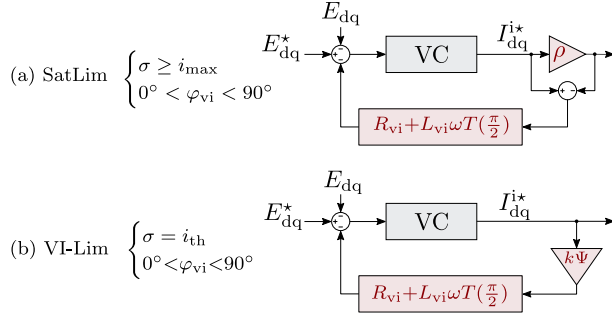


Fig. 3. From the unified current limiter framework, we can obtain the (a) SatLim, or (b) VI-Lim, by tuning σ and φ_{vi} .

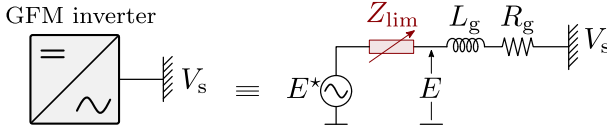


Fig. 4. Equivalent circuit representation of a GFM inverter during current limiting. The filter capacitor has been omitted for simplicity.

under all circumstances, according to (4). Further notice that setting $\varphi_{vi} = 0^\circ$ disables the VI inductance and yields a classical anti-windup gain, $R_{vi} = (k_p^{vc})^{-1}$; however, leveraging the VI in the anti-windup feedback loop of the SatLim allows for more control over the output current-phasor angle [30]. If we set $\sigma = i_{th}$ and $0^\circ < \varphi_{vi} < 90^\circ$, we retrieve a threshold VI-Lim, as shown in Fig. 3(b). Namely, Ψ starts increasing as soon as $\rho |I_{dq}^*| \geq i_{th}$, which is set lower than the maximum rated current, i_{max} . Because we tune k such that i_{max} is never breached, ρ will always be 1, which leaves us with the traditional threshold VI-Lim implementation.

Similar to an SG, we recognize that the behavior of a GFM inverter can be captured by a voltage source (of which the amplitude and angle is governed by the GFM primary controller) behind an impedance (the output filter impedance), as shown in Fig. 4. However, when the inverter output current is curtailed by the limiter—irrespective of what type of limiter is leveraged—the capacitor voltage, E , can no longer track the reference voltage, E^* , and the inverter output impedance increases. As such, a current limiter can be represented as a variable output impedance, Z_{lim} (see Fig. 4), of which the magnitude and angle depend on the current-limiter design and the present grid conditions. Further, we recognize that, during quasi-steady-state conditions, the fast inner controls can be assumed to be operating in steady state—the error feeding into the voltage and current controller are zero. From the aforementioned observations, and considering the unified framework in Fig. 3, we can find an expression for the equivalent limiter impedance, Z_{lim} , for quasi-steady-state conditions:

$$\begin{cases} \text{SatLim:} & Z_{lim} = \frac{(1-\rho)}{\rho} Z_{vi} \rightarrow \varphi_{lim} = \varphi_{vi}, \\ \text{VI-Lim:} & Z_{lim} = k\Psi Z_{vi} \rightarrow \varphi_{lim} = \varphi_{vi}. \end{cases} \quad (5)$$

Notice in (5), the angle of the equivalent-limiter impedance, φ_{lim} , is identical to the impedance angle of the limiter virtual

impedance, φ_{vi} . Understanding how the output impedance of a GFM inverter behaves during disturbances in the grid is critical for a variety of network-wide attributes and systems, such as network-wide stability, voltage support during faults, and power system protection [13].

In the following section, we utilize the unified current-limiting model and the equivalent circuit impedance of the limiters to compute the power-angle characteristics which serves as a tool to assess the transient stability margins of GFM inverters.

IV. THEORETICAL ANALYSIS OF UNIFIED MODEL AND LARGE-SIGNAL STABILITY

Following the conceptual introduction of the unified framework presented in Section III, this section establishes theoretical models to provide deeper insights into how various current limiter types impact the transient-stability characteristics of GFM inverters and how these dynamics interact with different grid conditions. In this section, we first refresh the classic concepts of large-signal stability analysis rooted in traditional power systems theory, as we continue to use these well-established methods to analyze GFM-inverter stability boundaries. Then, we present a numerical model of the unified framework as a benchmark, followed by an analytical model for in-depth analysis.

A. Large-Signal Stability With Synchronous Generators

In its most simplified form, an SG can be represented as a voltage source, E , behind an impedance, Z_{SG} . Assuming a purely inductive internal impedance, the SG active output power can be expressed as a function of the rotor angle using the well-known power-angle equation [31]:

$$p = \frac{|V_s| |E|}{|Z_{SG}|} \sin(\delta), \quad (6)$$

where E denotes the no-load excitation voltage; V_s denotes the terminal voltage; Z_{SG} represents the internal impedance of the SG with an impedance angle, φ_{SG} ; and δ denotes the angle between the voltage phasors, E and V_s ($\delta = \angle E - \angle V_s$). The internal rotor angle of the SG follows the droop equations in (1), where, in the context of SGs, θ denotes the instantaneous rotor angle. By leveraging (6) and (1), the rotor stability of SGs can be assessed. Fig. 5 illustrates the power-angle curve from (6) for $|V_s| = 1$ pu and $|V_s| = 0.5$ pu, which represents a grid voltage drop caused by a disturbance or fault. Consider the scenario where the SG is operating at the equilibrium point ‘a’ with a power reference of $p^* = 1$ pu when a sudden voltage drop to 0.5 pu occurs in the grid. The output power will follow a trajectory ‘a-b-c-d’ as shown in Fig. 5, where we assume the fault is cleared at point ‘c’. To simplify the discussion, no inertia is considered in Fig. 5. The mechanical inertia of the SG can lead to angle (δ) overshoot and oscillations during and after the disturbance, which in turn can reduce the transient stability margins of the generator; however, the principle behind the trajectory on the power-angle curves, and the position of equilibrium points remains valid.

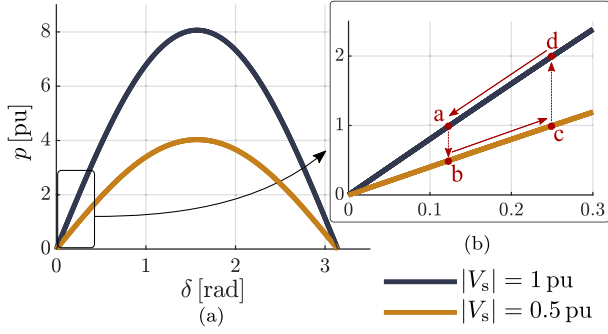


Fig. 5. (a) Power-angle curve of a generic SG for different grid voltages, V_s . (b) Conceptual trajectory of the active output power and rotor angle, δ , during a generic voltage drop (no inertia response considered).

Based on the interpretations of (6) and (1), we can extend this analysis to assess the transient stability of GFM inverters; however, to do so, we must modify the interpretation of a power-angle curve. For SGs, the $p(\delta)$ curves map the output power as a function of the rotor angle relative to the grid-voltage phasor angle [31]. In contrast, for GFM inverters, which do not have a physical rotor, we refer to the primary-controller internal angle, θ . We denote the angle $\delta' = \theta - \angle V_g$. We use the notation δ' to differentiate from the classical interpretation of δ with SGs.

B. Numerical Model of the Unified Framework

In this subsection, we leverage the unified current-limiting framework to establish a system of continuous nonlinear algebraic equations from which we can compute the $p(\delta')$ curves for any of the two limiter types. This numerical model serves as a benchmark for our analytical model and stability analysis.

Power-angle characteristics can only capture quasi-steady-state characteristics of the inverter; therefore, we can once more assume the fast inner-controls to be in steady state, and the error signals feeding into the voltage and current controller to be zero. The assumptions, starting from the equivalent-network representation of a GFM inverter in Fig. 4, lead to:

$$\begin{aligned} 0 &= E^* - E - V_{vi} \\ &= E^* - \rho I^* Z_g - V_s - (1 - \rho + \rho k \Psi) I^* Z_{vi}, \end{aligned} \quad (7)$$

where V_{vi} denotes the voltage drop across the VI of the unified current limiter; Z_g denotes the grid-side inductive output-filter impedance; ρI^* denotes the output current of the inverter, for which we make the approximation that $\rho I^* = I_i = I_g$, i.e., we neglect the LCL filter capacitor current for simplicity, though adding the capacitor into the model adds only one branch and can straightforwardly be integrated into the present model [32]; and ρ and Ψ are discontinuous functions as defined in (3) and (4), respectively. For a system of nonlinear algebraic equations to be solvable, we rewrite the discontinuous functions of (3) and (4) into continuous forms using the smooth maximum unit [33]:

$$\rho = -\frac{1}{2} \left(-1 - \frac{i_{\max}}{|I^*|} + \sqrt{\left(\frac{i_{\max}}{|I^*|} - 1 \right)^2} \right), \quad (8)$$

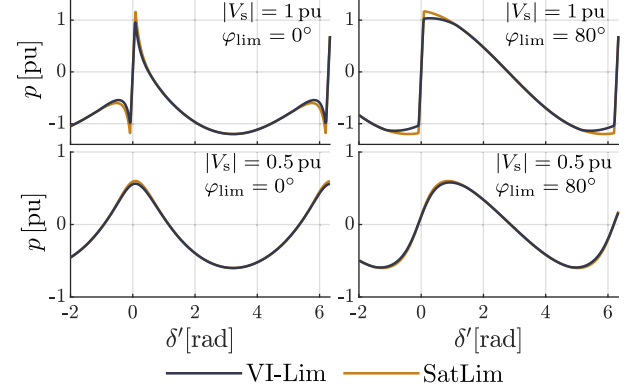


Fig. 6. Numerically computed $p(\delta')$ curves for the two considered current limiters when (top) $|V_s| = 1$ pu and (bottom) $|V_s| = 0.5$ pu. The angle of the VI in the VI-Lim and the SatLim is set to 0° (left) and 80° (right).

$$\Psi = \frac{1}{2} \left(\frac{\rho |I^*| - \sigma}{i_{\max} - i_{th}} + \sqrt{\left(\frac{\rho |I^*| - \sigma}{i_{\max} - i_{th}} \right)^2} \right), \quad (9)$$

where we substitute (8) and (9) in (7), respectively. From (7), (8), and (9), along with the droop equations that govern the voltage reference (1)–(2), we derive the following system of nonlinear algebraic equations. Because (7) is a complex equation, we split (7) into a real and imaginary part, denoted by the subscripts ‘d’ and ‘q’, respectively:

$$\begin{cases} 0 = E_{dq}^* - \rho(R_g + T(\frac{\pi}{2})\omega_s L_g)I_{dq}^* - V_{s,dq} \\ \quad - (1 - \rho + \rho k \Psi)(R_{vi} + T(\frac{\pi}{2})\omega_s L_{vi})I_{dq}^*, \\ 0 = -E_{dq}^* + (e_0 + m_q(q^* - q))T(-\frac{\pi}{2})e_1, \\ 0 = -E_{dq} + V_{s,dq} + \rho(R_g + T(\frac{\pi}{2})\omega_s L_g)I_{dq}^*, \\ 0 = -q + \rho(I_{dq}^* \times E_{dq}) \end{cases} \quad (10)$$

More details on how to setup this continuous algebraic system of equations, while incorporating the filter capacitor is given in [32]. From (10), we compute the output power for a given δ' as:

$$p = \rho E_{dq}^T I_{dq}^*. \quad (11)$$

We retrieve $p(\delta')$ by computing (11) while sweeping for δ' . Fig. 6 shows the resulting $p(\delta')$ curves for the two current limiters while the grid voltage is operating at $|V_s| = 1$ pu (top graphs) and when considering a voltage drop at the inverter terminals of 0.5 pu (bottom graphs). The curves are computed for two cases: $\varphi_{vi} = \varphi_{lim} = 0^\circ$ and $\varphi_{vi} = \varphi_{lim} = 80^\circ$. (We retrieve from (5) that $\varphi_{lim} = \varphi_{vi}$.) Here, the maximum rated current is set to $i_{\max} = 1.2$ pu, and the threshold current for the VI-Lim is set to $i_{th} = 1.1$ pu. Notice that the $p(\delta')$ curves for the SatLim and the VI-Lim are nearly identical. Only a slight difference in output power between the SatLim and VI-Lim remains during light overcurrent conditions: VI-Lim is inherently designed to only hit $i_{\max}$ in the worst case scenario, which translates to an under-usage of the overcurrent capabilities during all other

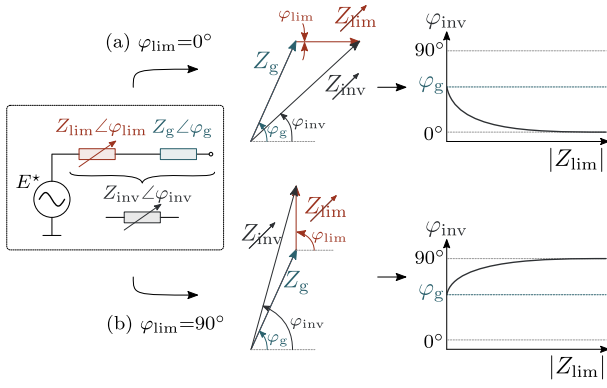


Fig. 7. Conceptual impedance phasor diagrams for GFM inverters: (a) one with Lim⁰ and (b) with Lim⁹⁰. For severe limiting, the inverter impedance asymptotically nears the equivalent limiter impedance angle.

current-limiting conditions that are not worst case. If we neglect this small difference in the output power between the VI-Lim and the SatLim, we conclude that the $p(\delta')$ curves of the unified current-limiter model are solely defined by the equivalent current limiter impedance angle, φ_{lim} . Based on this equivalence, in the remainder of this work, we refer to limiters by their respective equivalent-network impedance angle, φ_{lim} , rather than calling upon the individual limiter types, and we denote them by ‘Lim $^{\varphi_{\text{lim}}}$ ’. For example, the limiter Lim⁰ could be the SatLim with $\varphi_{\text{vi}} = 0^\circ$, or the VI-Lim with $\varphi_{\text{vi}} = 0^\circ$, and both will have nearly identical $p(\delta')$ characteristics. *Remark:* Although the $p(\delta')$ curves for SatLim and VI-Lim are nearly identical, one must be mindful that the $p(\delta')$ curves only capture quasi-static behaviors, and do not capture the small-signal response of current limiters.

This numerical model sets the benchmark to compute $p(\delta')$ curves which can be leveraged to assess the large-signal stability of GFM inverters; however, it might not provide insights into how these unconventional curves are formulated or what drives the behavior of such characteristics. The following derivation of the analytical model provides an answer to these questions.

C. Analytical Model of $p(\delta')$ Characteristics

In this subsection, we derive an analytical expression for the $p(\delta')$ curves by taking an intuitive approach to understanding the mechanisms driving the GFM inverter $p(\delta')$ behavior. We start off by examining two extreme cases: limiters with an equivalent-network impedance angle, $\varphi_{\text{lim}} = 0^\circ$, and limiters with $\varphi_{\text{lim}} = 90^\circ$. Assuming the internal inverter reference voltage magnitude, $|E^*|$, to remain close to the nominal value, 1 pu, we represent the GFM controls as a controlled voltage source, E^* , behind an impedance, Z_g . The current limiter appears as a variable impedance, Z_{lim} , which, lumped with Z_g , forms the total inverter output impedance, Z_{inv} , as shown in Fig. 7. For severe limiting conditions, Fig. 7 illustrates that the current-limiter impedance, Z_{lim} , dominates the total inverter output impedance such that $Z_{\text{inv}} \approx Z_{\text{lim}}$. In other words, for a current limiter with $\varphi_{\text{lim}} = 0^\circ$, the inverter behaves more resistively under current limiting (Fig. 7(a)), while for limiters with $\varphi_{\text{lim}} = 90^\circ$,

it behaves more inductively (Fig. 7(b)). From Fig. 7, we can now understand how the GFM inverter output power, p , behaves as a function of δ' under these different current-limiting conditions, which is illustrated in Fig. 8. For the Lim⁰ GFM inverter in Fig. 8(b), the inverter output impedance becomes increasingly resistive as the limiter engages more forcefully (see Fig. 7(a)), causing the output power to drop rapidly along the δ' axis. In contrast, in case of the Lim⁹⁰ GFM inverter in Fig. 8(c), the inverter output impedance becomes increasingly inductive as the limiter engages more forcefully (see Fig. 7(b)), which translates into an output power that remains near rated values longer along the δ' axis.

Leveraging these insights, we now derive an analytical expression to describe $p(\delta')$. To simplify the analytical derivation, we make the assumption that $|E^*| = 1$ pu under all circumstances; namely, from (2), considering typical values for $m_q = 1\% - 5\%$, and knowing that the maximum reactive output power is curtailed, this assumption is reasonable. We illustrate the validity of this assumption later in this section by comparing the analytical and numerical results.

Based on the equivalent circuit depicted in Fig. 8(a), we can write the GFM inverter output power as:

$$S = E \cdot \bar{I} = \left(V_s + (E^* - V_s) \cdot \frac{Z_g}{Z_{\text{inv}}} \right) \cdot \left(\frac{E^* - V_s}{Z_{\text{inv}}} \right), \quad (12)$$

After some algebraic manipulations, and recognizing that $Z_{\text{inv}} = Z_{\text{lim}} + Z_g$, we get

$$S = \frac{1}{|Z_{\text{inv}}|^2} (|E^*|^2 Z_g - |V_s|^2 Z_{\text{lim}} - E^* \bar{V}_s Z_g + \bar{E}^* V_s Z_{\text{lim}}),$$

from which we retrieve the active power:

$$p = \text{Re}\{S\} = \frac{|E^*|^2}{|Z_{\text{inv}}|^2} \{ |Z_g| \cos \varphi_g - \frac{|V_s|^2}{|E^*|^2} |Z_{\text{lim}}| \cos \varphi_{\text{lim}} - \frac{|V_s|}{|E^*|} |Z_g| \cos(\delta' + \varphi_g) + \frac{|V_s|}{|E^*|} |Z_{\text{lim}}| \cos(\varphi_{\text{lim}} - \delta') \}. \quad (13)$$

Defining $k_f = \frac{|V_s|}{|E^*|}$ as a grid voltage ratio coefficient, we simplify (13) to:

$$p = \frac{|E^*|^2}{|Z_{\text{inv}}|^2} \{ |Z_g| (\cos \varphi_g - k_f \cos(\delta' + \varphi_g)) - k_f |Z_{\text{lim}}| (k_f \cos \varphi_{\text{lim}} - \cos(\delta' - \varphi_{\text{lim}})) \}, \quad (14)$$

where $|Z_{\text{inv}}|$ and $|Z_{\text{lim}}|$ are both functions of δ' ; during normal operation, the limiter is disabled, leaving $Z_{\text{inv}} = Z_g$, $Z_{\text{lim}} = 0$; under limiting conditions, the output current of the inverter saturates at the maximum rated value, independent of terminal voltage. Thus, the equivalent inverter impedance magnitude is:

$$|Z_{\text{inv}}| = \max \left(|Z_g|, \frac{|\Delta E|}{i_{\text{max}}} \right), \quad (15)$$

where $|\Delta E| = |E^* - V_s|$ is once more a function of δ' , and can be determined using the cosine law:

$$|\Delta E| = |E^*| \sqrt{1 + k_f^2 - 2k_f \cos \delta'}. \quad (16)$$

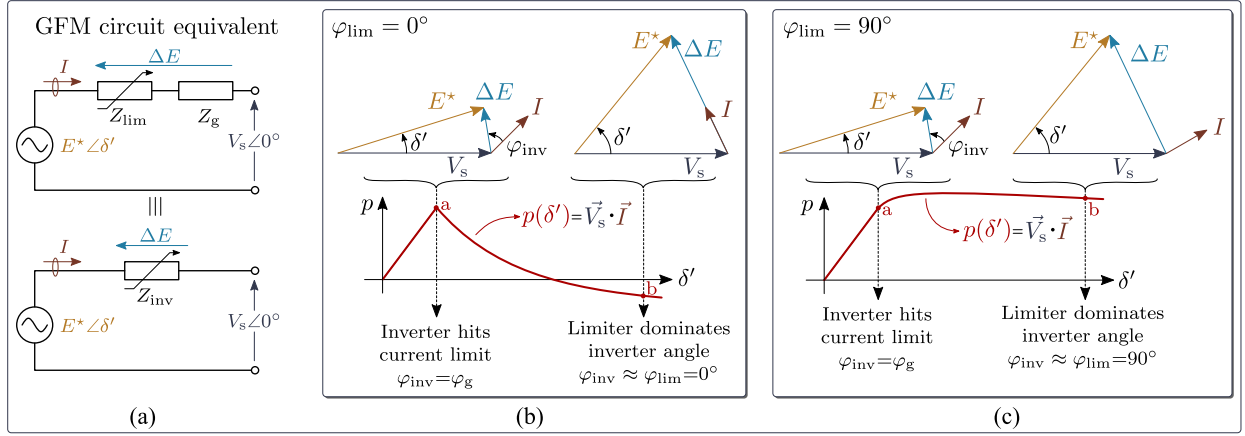


Fig. 8. Graphical explanation of how the $p(\delta')$ characteristics are manipulated for the different current limiters: (a) the phasor impedance diagram of the GFM inverter, (b) the voltage-current phasor diagrams along with the $p(\delta')$ relation for the Lim^0 GFM inverter, and (c) a Lim^{90} GFM inverter.

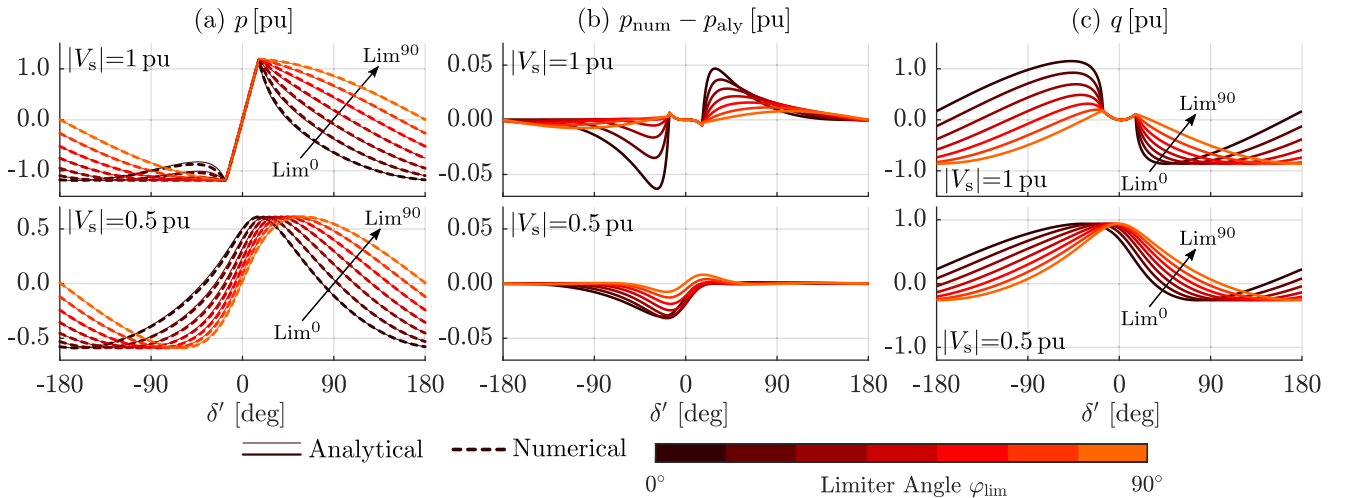


Fig. 9. (a) $p(\delta')$ curves with different φ_{lim} when $|V_s| = 1.0$ pu (top) and $|V_s| = 0.5$ pu (bottom); (b) Differences between numerical and analytical $p(\delta')$ curves with different φ_{lim} when $|V_s| = 1.0$ pu (top) and $|V_s| = 0.5$ pu (bottom); (c) $q - \delta'$ curves with different φ_{lim} when $|V_s| = 1.0$ pu (top) and $|V_s| = 0.5$ pu (bottom).

Leveraging the cosine law, we can express the limiter impedance as:

$$|Z_{\text{lim}}| = -|Z_g| \cos(\varphi_g - \varphi_{\text{lim}}) + \sqrt{|Z_{\text{inv}}|^2 - |Z_g|^2 \sin^2(\varphi_g - \varphi_{\text{lim}})}. \quad (17)$$

Substituting (16) in (15), (15) in (17) and (14), and (17) in (14) finally delivers an analytical expression for the active output power as a function of δ' . For analysis, (14) can be decomposed into three components, with each component influencing the active power injection:

$$\Gamma = \frac{|E^*|}{|Z_{\text{inv}}|}, \quad (18)$$

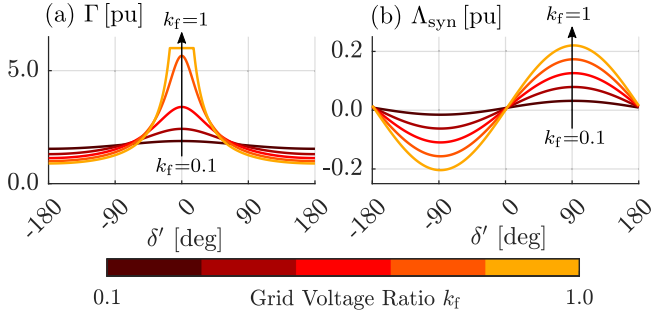
$$\Lambda_{\text{syn}} = |Z_g|(\cos \varphi_g - k_f \cos(\delta' + \varphi_g)), \quad (19)$$

$$\Lambda_{\text{lim}} = k_f |Z_{\text{lim}}| (k_f \cos \varphi_{\text{lim}} - \cos(\delta' - \varphi_{\text{lim}})). \quad (20)$$

Substituting (18), (19), and (20) back into (14), yields a concise analytical expression for the output power:

$$p = \Gamma^2 (\Lambda_{\text{syn}} - \Lambda_{\text{lim}}). \quad (21)$$

Fig. 9(a) depicts the matching $p(\delta')$ curves of the analytical and numerical model for increasing values of the current-limiter impedance angle, φ_{lim} , ranging from $0^\circ - 90^\circ$, and thereby validating the analytical derivation of $p(\delta')$. Fig. 9(b) illustrates the difference between the numerical and analytical $p(\delta')$ characteristics. Notice the maximum error between the two models remains within ± 0.06 pu. For $\varphi_{\text{lim}} = 90^\circ$, the peak error decreases to ± 0.01 pu due to better power support. This error is attributed to the assumption of a constant $|E^*| = e^* = 1$ pu in the analytical model. We can clarify this as follows: The reference-voltage magnitude, e^* , in the numerical model follows the droop equation in (2). Consider $q^* = 0$ pu, $e_0 = 1$ pu, $m_q = 0.05$ pu, and $i_{\text{max}} = 1.2$ pu, which are default settings

Fig. 10. (a) $\Gamma - \delta'$ curves and (b) $\Lambda_{\text{syn}} - \delta'$ curves for different k_f .

throughout this paper, then we recognize that:

$$\begin{aligned} e_{\min, \max}^* &= e_0 + m_q(0 \pm q_{\max}) = e_0 \pm m_q e_0 i_{\max} \\ &= e_0 \pm 0.06 \text{ pu.} \end{aligned} \quad (22)$$

In other words, in the most severe case, the difference ($e_{\text{aly}}^* - e_{\text{num}}^*$), and by extension ($p_{\text{aly}}^* - p_{\text{num}}^*$), cannot surpass ± 0.06 pu (subscripts aly and num denote their respective values from analytical and numerical models). Hence, we conclude that fixing $e_{\text{aly}}^* = e_0$ is reasonable, yielding acceptable errors even under worst-case fault scenarios. Although for different parameter choices of q^* , m_q , and $i_{\max}$, this error can vary, under reasonable practical constraints, this variation will remain minimal.

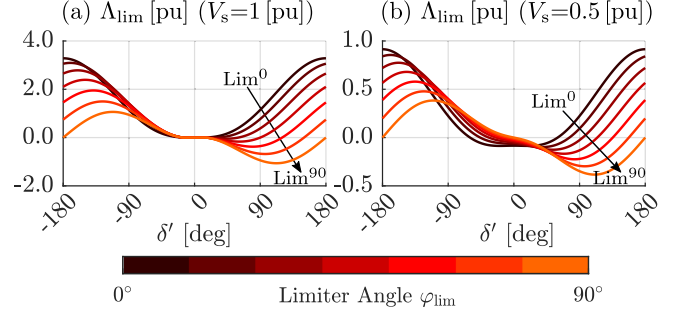
In a similar fashion as outlined above, we can compile an analytical expression for the reactive output power, $q(\delta')$, by taking the imaginary part of (12):

$$\begin{aligned} q &= \frac{|E^*|^2}{|Z_{\text{inv}}|^2} \{ |Z_g| (\sin \varphi_g + k_f \sin(\delta' + \varphi_g)) \\ &\quad - k_f |Z_{\text{lim}}| (k_f \sin \varphi_{\text{lim}} - \sin(\delta' - \varphi_{\text{lim}})) \}. \end{aligned} \quad (23)$$

Fig. 9(c) portrays the $q(\delta')$ curves for various φ_{lim} and grid voltages. Notice the Lim^{90} provides significantly higher reactive power support in the early stage of current limiting, which translates to better voltage support during grid faults.

1) *Virtual Short-Circuit Current Γ* : From (18), it is evident that Γ represents a virtual short-circuit power as a function of δ' under current-constrained conditions ($|Z_{\text{inv}}|$ is a function of δ'). Fig. 10(a) illustrates the $\Gamma - \delta'$ curves for various grid voltage ratios, k_f . During normal operation, when $Z_{\text{inv}} = Z_g$, Γ is a constant value independent of δ' , with the short-circuit power curtailed only by the value of the output filter impedance, Z_g (this coincides with $\Gamma = 6$ pu in Fig. 10(a)). Under current-limiting conditions, the inverter output impedance, $|Z_{\text{inv}}|$, increases due to the current-limiting action, which in turn, reduces the virtual short-circuit power of the inverter, Γ . More severe grid-voltage drops—denoted by k_f —increase the engagement of the limiter, which further reduces the virtual GFM inverter short-circuit power, as illustrated in Fig. 10(a).

2) *Synchronous Machine Component Λ_{syn}* : Λ_{syn} , as introduced in (19), is influenced by the grid-side filter impedance, Z_g , and the grid-voltage ratio, k_f . As such, Λ_{syn} is unrelated to current limiting, and therefore reflects the inherent characteristics of the GFM inverter under non current-limiting conditions, similar to

Fig. 11. $\Lambda_{\text{lim}} - \delta'$ curves with different φ_{lim} when $|V_s| = 1.0$ pu (a) and $|V_s| = 0.5$ pu (b).

a synchronous generator. Namely, if in (14) we set $|Z_{\text{lim}}| = 0$, which implies that $Z_{\text{inv}} = Z_g$ and $|\Lambda_{\text{lim}}| = 0$, we get:

$$p = \Gamma^2 \Lambda_{\text{syn}} = \frac{|E^*|^2}{|Z_g|} \cos \varphi_g - \frac{|E^*| |V_s|}{|Z_g|} \cos(\delta' + \varphi_g),$$

which is analogous to the well-known power-angle equation of SGs or constant voltage source with a fixed output impedance. Fig. 10(b) displays the $\Lambda_{\text{syn}} - \delta'$ curve for various grid voltage ratios, k_f . The parallels between Figs. 5 and 10(b) further support that Λ_{syn} embodies the ‘synchronous-machine-like’ behaviors of the GFM inverter.

3) *Current-Limiter Component Λ_{lim}* : In (20) and (21), the current limiter component, Λ_{lim} , represents the reduction in GFM inverter power injection due to current limiting. Fig. 11 illustrates the $\Lambda_{\text{lim}} - \delta'$ curves for different values of φ_{lim} at (a) $|V_s| = 1.0$ pu and (b) $|V_s| = 0.5$ pu.

Λ_{lim} is influenced by two factors: the gain $k_f |Z_{\text{lim}}|$ and a sum of cosine terms, $k_f \cos \varphi_{\text{lim}} - \cos(\delta' + \varphi_{\text{lim}})$. The primary factors influencing the gain $k_f |Z_{\text{lim}}|$ are the grid voltage difference, ΔE , the grid-side filter impedance, Z_g , and the maximum current limit, $i_{\max}$. Since Z_g and $i_{\max}$ are fixed values, $k_f |Z_{\text{lim}}|$ acts as a scaling factor that influences the extent of power reduction as a function of the grid voltage. Second, the term $k_f \cos \varphi_{\text{lim}} - \cos(\delta' + \varphi_{\text{lim}})$ is influenced by the grid voltage, the limiter impedance angle, φ_{lim} , and δ' . Fig. 11 illustrates that more inductive settings for φ_{lim} reduces Λ_{lim} across the δ' spectrum. In other words, an inductive current limiter leads to more active power support under limiting conditions than a resistive current limiter.

4) *Influence of Grid Strength*: Lastly, we can leverage the unified current-limiter model to analyze the influence of the grid strength on the power-angle behavior of the GFM inverter. Figs. 12 and 13 display the $p(\delta')$ curves for various grid impedance magnitudes ($|Z_\ell|$) and angles ($|\varphi_\ell|$), respectively. Fig. 12 shows that an increase in grid impedance—i.e., a weaker grid condition—allows for more angle swing to achieve the same power injection. This translates to a larger unstable-equilibrium angle, δ'_{u} , during current limiting. In other words, a weaker grid condition can aid the GFM inverter to remain grid-synchronized during current limiting. This effect is more pronounced for the Lim^0 in Fig. 12(a) where the stable operating region is much smaller than the Lim^{80} in Fig. 12(b). Fig. 13 shows that the

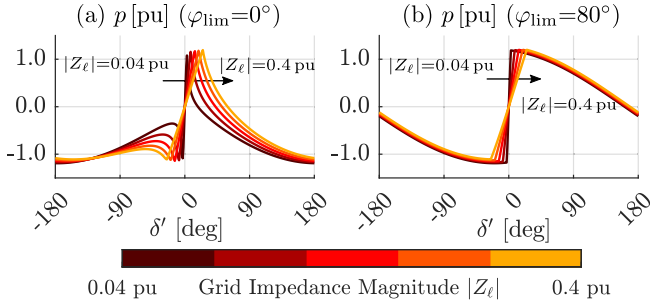


Fig. 12. $p(\delta')$ curves with different $|Z_g|$ ($|V_s| = 1.0$ pu) when $\varphi_{\text{lim}} = 0^\circ$ (a) and $\varphi_{\text{lim}} = 80^\circ$ (b).

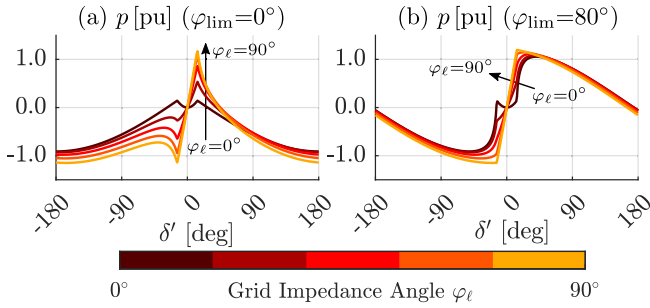


Fig. 13. $p(\delta')$ curves with different φ_g ($|V_s| = 1.0$ pu): when (a) $\varphi_{\text{lim}} = 0^\circ$ and (b) $\varphi_{\text{lim}} = 80^\circ$.

grid-impedance angle has a significant influence on the active power injection of the GFM inverter during current limiting. For predominantly resistive grids, $p(\delta')$ and $q - V$ are no longer decoupled, i.e., a δ' increase doesn't drive a greater active power injection. In these resistive-grid scenarios, Fig. 13(b) illustrates that an inductive current limiter can aid the active power injection during limiting; the 'virtually' created inductiveness by the limiter compensates for the lack of inductiveness in the grid. This further supports the importance of GFM inverters with inductive current limiters in maintaining grid stability.

In summary, this section illustrated how the unified current-limiter model is set up, and how it enables the derivation of numerical and analytical power-angle characteristics of current-limited GFM inverters. Deriving the analytical power-angle curves illustrated that the behavior of current-limited GFM inverters is influenced by three parameters: i) the virtual short-circuit current, ii) the synchronous-machine component, and iii) the GFM current-limiter component. Each of these attributes illustrates key features of GFM inverters under current constraints in a clear and concise manner. A sensitivity analysis leveraging the analytical power-angle characteristics showed that a more inductive current limiter increases the overall inductiveness of the inverter output impedance, which: i) increases the stability margins of the inverter, ii) increases active power support during current limiting under angle-related conditions, and iii) improves grid-voltage support during voltage drops.

How all the aforementioned parameters and power-angle behaviors eventually influence the transient stability margins of GFM inverter is investigated in the next section.

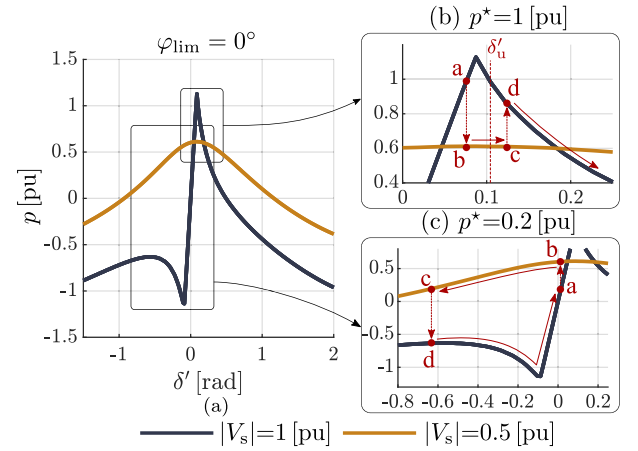


Fig. 14. $p(\delta')$ curves for a Lim^0 GFM inverter under different grid voltage conditions, along with the conceptual $p(\delta')$ trajectories for a generic voltage drop with (b) $p^* = 1$ pu and (c) $p^* = 0.2$ pu.

V. LARGE-SIGNAL STABILITY ANALYSIS OF GFM INVERTERS AND IBR-DOMINANT POWER SYSTEMS

Leveraging the derived $p(\delta')$ characteristics, we can analyze the large-signal stability (i.e., transient stability) of GFM inverters and inverter-dominated power systems in a similar manner as with SGs. Analyzing and mapping the transient-stability margins of GFM inverters provides insight into how robust a GFM-dominated power system is during transients and disturbances. Similar to classical transient-stability approaches, we start from a single inverter connected to an infinite bus setup and expand to investigating the stability of a multi-generator inverter-driven power grid.

We first consider a generic grid voltage drop to 0.5 pu and conceptually explain the differences in the power-angle dynamics of GFM inverters when employing either a Lim^0 (i.e., purely resistive) or Lim^{90} (i.e., purely inductive) limiter. (Though inertia response in the primary controller is not considered, the following outlined principles remain the same.) Through simulations, we then assess the GFM transient stability under three generic fault types: voltage sag, phase jump, and frequency shift. Notice that these fault types, or a combination of those, constitute all realistic faults that can occur in a grid (e.g., a line fault is a combination of a voltage drop, phase shift, and potentially followed by a frequency swing). Following, we validate the device-level results through a hardware experiment. Lastly, we present simulation results of a multi-inverter 14-bus network that is subjected to realistic unbalanced line faults and illustrate system-level transient stability and unbalanced voltage support characteristics of the interconnected GFM inverters using the proposed unified model.

A. Lim^0 GFM Inverter

Fig. 14 shows the $p(\delta')$ curves for an inertia-less GFM inverter with a Lim^0 . The dark blue curve represents nominal grid-voltage conditions ($|V_s| = 1$ pu), while the yellow curve represents a degraded grid voltage ($|V_s| = 0.5$ pu). In Fig. 14(b) where $p^* = 1$ pu, the GFM inverter operates in the pre-fault

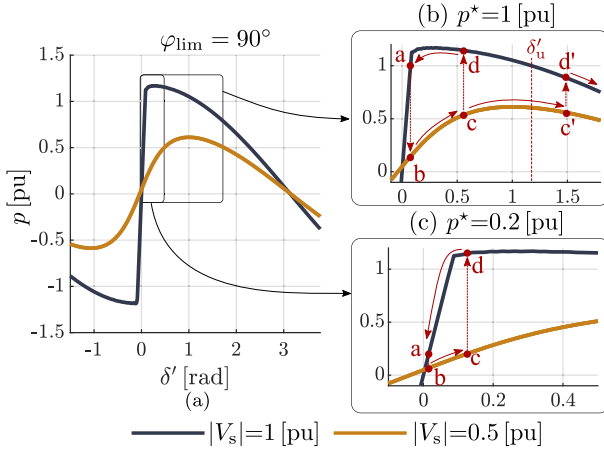


Fig. 15. $p(\delta')$ curves for a Lim^{90} GFM inverter under different grid voltage conditions (a), along with the conceptual $p(\delta')$ trajectories for a generic voltage drop with (b) $p^* = 1$ pu and (c) $p^* = 0.2$ pu.

stable equilibrium point 'a'. The voltage drop is initiated, causing the inverter output power to drop to point 'b'. Because $p_b < p^*$, δ' starts increasing as per (1), and the operating point shifts along the yellow curve. At operating point 'c', the fault is cleared, and the inverter output power jumps to point 'd'. Because δ' has exceeded the unstable equilibrium point, δ'_u , the inverter becomes transient unstable. The inverter will only return to equilibrium if δ' never surpasses δ'_u (both during and post fault). In Fig. 14(c), with $p^* = 0.2$ pu, the voltage drop causes the inverter to suddenly increase its output power and jumps from the pre-fault stable equilibrium point 'a' to point 'b'. Because $p_b > p^*$, the GFM inverter frequency and angle decelerate as per (1). At point 'c', the inverter reaches a new stable equilibrium. After the voltage drop clears, the inverter jumps to point 'd', returning to the prefault equilibrium point 'a', all the while drawing significant active power from the grid which is not desired during a fragile post-fault recovery process. Both cases shown in Fig. 14 illustrate that the Lim^0 GFM inverter has stability drawbacks during large disturbances.

B. Lim^{90} GFM Inverter

Fig. 15 shows the $p(\delta')$ curves for a GFM inverter with a Lim^{90} for $|V_s| = 1$ pu (dark blue) and $|V_s| = 0.5$ pu (yellow). In Fig. 15(b), in normal operation with $p^* = 1$ pu, the GFM inverter operates at the stable equilibrium point 'a'. Once a voltage drop initiates, the inverter output power jumps to point 'b'. Because $p_b < p^*$, the inverter reference angle starts increasing, and the operating point shifts along the yellow curve. When the voltage drop clears at point 'c', the inverter jumps to point 'd' where the current is still being curtailed; however, δ' did not exceed the unstable equilibrium point (note δ'_u is significantly greater than for the Lim^0), allowing the inverter to return to the pre-fault stable equilibrium point, 'a'. If the voltage drop prolongs past δ'_u , the inverter will eventually become unstable. In Fig. 15(c), with $p^* = 0.2$ pu, the inverter reaches a stable equilibrium at point 'c'. Once the fault clears, the inverter jumps to point 'd', after which it can return to the pre-fault operating point without momentary

TABLE I
SIMULATION PARAMETERS

Parameter	Value	Unit	Parameter	Value	Unit
V_{base}	16	V	L_f	$120 \cdot 10^{-6}$	H
S_{base}	32.0	VA	R_f	$200 \cdot 10^{-3}$	Ω
I_{base}	1.15	A	C_f	$10 \cdot 10^{-6}$	F
k_p^{vc}	0.60	pu	L_g	$5 \cdot 10^{-3}$	H
k_i^{vc}	1010.64	pu	R_g	$40 \cdot 10^{-3}$	Ω
k_p^{cc}	0.06	pu	i_{max}	1.2	pu
k_i^{cc}	94.25	pu	m_p	2	%
p^*	1.0	pu	m_q	5	%
q^*	0	pu	ω_0	$60 \cdot 2\pi$	rad/s

negative power injection. Note that these insights obtained apply to any inner-control-loop-based limiter (i.e., either using current-control or voltage-control based) as long as the limiter angles are identical, based on the proposed unified model framework. In the following subsection, we illustrate this through full-order simulations.

C. Single-Inverter Fault Simulations

We validate the theoretical $p(\delta')$ curves and findings using full-order EMT simulations of a single-machine-infinite-bus (SMIB) scenario. We utilize the GFM inverter control system shown in Fig. 2, with parameters listed in Table I. We simulate three generic grid disturbance scenarios—a 0.5-pu voltage drop for 1 s, a 1-Hz frequency jump for 0.2 s, and a 30° phase jump—while considering Lim^0 and Lim^{75} SatLim and VI-Lim current limiters. The $p(\delta')$ trajectories from simulations are depicted as solid lines in Fig. 16, alongside theoretical curves from the unified analytical current-limiter model shown as dashed lines. Notice that the droop primary controller has filtering on the measured power, which induces a small inertia-like dynamic response during transients.

Fig. 16 shows that the simulation results align with the analytical curves. For increasing δ' , the output power of the Lim^0 GFM inverter (left graphs in Fig. 16) degrades much faster compared to the Lim^{75} GFM inverter (right graphs in Fig. 16) upon current limiting. This translates to significantly smaller stability margins for the Lim^0 compared to the Lim^{75} GFM inverter. Consequently, all three fault scenarios trigger instability for the Lim^0 GFM inverter, whereas the Lim^{75} GFM inverter can ride through all the faults and recover to pre-fault operation. Fig. 17 further illustrates these phenomena through time-domain plots of the phase portrait and the inverter output power. The solid lines represent Lim^{75} , and the dashed lines represent Lim^0 . The output-power plots indicate that the Lim^0 causes significant active power drops during limiting, prompting the droop laws to increase the GFM inverter frequency, which results in a significant increase in δ' . During and after fault clearance, the Lim^0 phase plots diverges due to a loss of synchronism, preventing direct restoration to the original steady state. Conversely, the Lim^{75} continues to inject active power during and after the fault, keeping the frequency and angle deviations relatively small, which enables the GFM inverter to return to the pre-fault steady state.

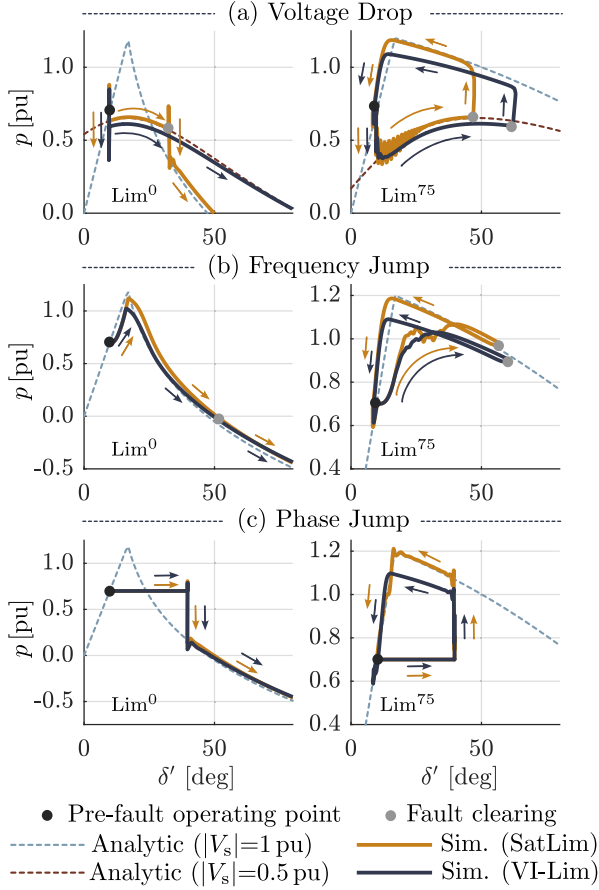


Fig. 16. Analytical and simulated $p(\delta')$ curves for (a) a 0.5-pu voltage drop for 1 s, (b) 1-Hz frequency jump for 0.2 s, and (c) a 30° phase jump, for SatLim and VI-Lim GFM inverters with Lim^0 (left) and Lim^{75} (right).

Notice in Fig. 16 the offset between the SatLim and VI-Lim output powers. This is attributed to the VI-Lim not being able to fully use the overcurrent capabilities of the inverter because limiting initiates as soon as $|I_{dq}^i| > i_{th}$. As illustrated in Fig. 6, SatLim and VI-Lim with the same impedance angle would have identical δ'_u , but due to the under-use of overcurrent capability, the VI-Lim angle diverges more, yielding shorter critical clearing time (not shown), i.e., less stability margin. Overall, the simulation results align with the analyses presented, which confirms the validity of the unified model and demonstrates exemplary use of them.

D. Single-Inverter Hardware Experiment

To validate the unified model proposed in this paper, we use a hardware test bed forming a SMIB setup running two case studies. The test cases impose two typical grid contingencies: (i) a grid-side voltage frequency jump from 60 Hz to 59.4 Hz for 0.15 s, and (ii) a grid-side voltage magnitude drop from 1.0 pu to 0.5 pu for 0.5 s. In each case study, both the VI-Lim and SatLim configurations are implemented and tested with different virtual impedance angles (0° and 50°). Figs. 18 and 19 illustrate the hardware experimental setup, and Table II shows the inverter hardware and control parameters.

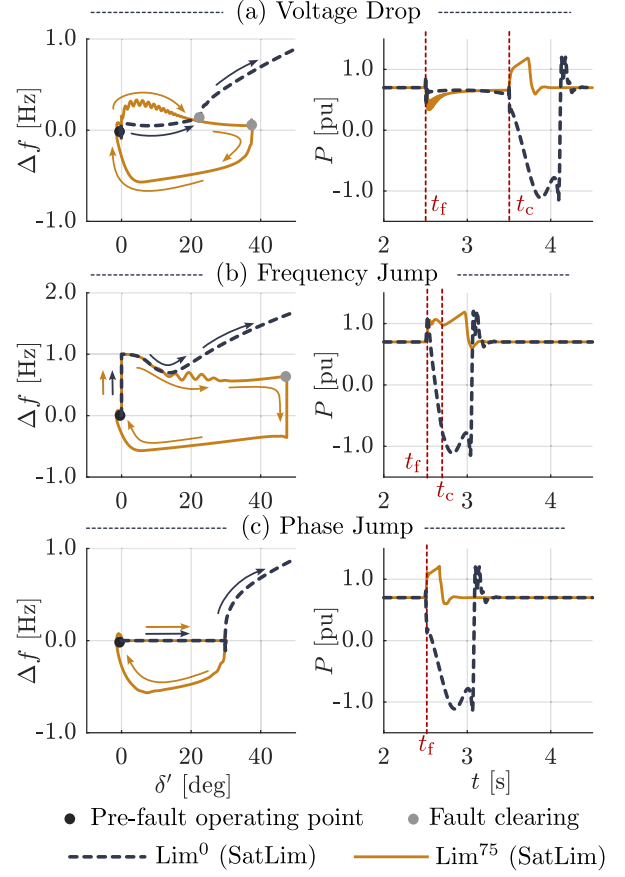


Fig. 17. Simulated Δf - $\Delta\delta'$ ($\Delta f = f_{\text{GFM}} - f_g$ and $\Delta\delta' = \delta' - \delta'_0$) and $P(t)$ curves for (a) a 0.5-pu voltage drop, (b) 1-Hz frequency jump, and (c) a 30° phase jump, for Lim^0 and Lim^{75} SatLim GFM inverters.

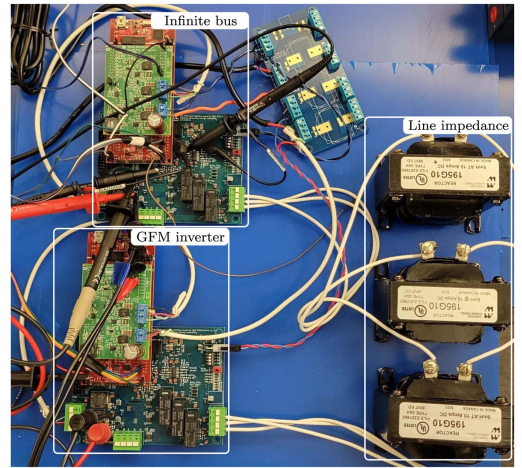


Fig. 18. SMIB hardware test bed for experimental validation.

Fig. 20 presents the experimental results with measured GFM inverter output line-to-line voltages and line currents, calculated powers (active, reactive, and apparent output powers, respectively, denoted by $S \in \mathbb{R}_{\geq 0}$, $P \in \mathbb{R}$, and $Q \in \mathbb{R}$), and the anti-windup/virtual impedance voltage magnitude fed to the voltage controller via the digital-to-analog converter of the

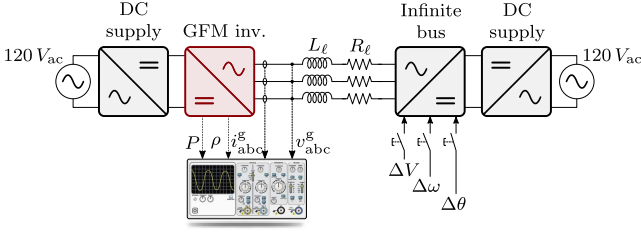


Fig. 19. Schematic of the hardware test bed with grid disturbance implemented through a programmable inverter and measurements illustrated.

TABLE II
HARDWARE EXPERIMENT PARAMETERS

Parameter	Value	Unit	Parameter	Value	Unit
V_{base}	16	V	L_f	$120 \cdot 10^{-6}$	H
S_{base}	32.0	VA	R_f	$200 \cdot 10^{-3}$	Ω
I_{base}	1.15	A	C_f	$10 \cdot 10^{-6}$	F
k_p^{vc}	0.60	pu	L_g	$5 \cdot 10^{-3}$	H
k_i^{vc}	1010.64	pu	R_g	$40 \cdot 10^{-3}$	Ω
k_p^{cc}	0.06	pu	$i_{max,rms}$	1.2	pu
k_i^{cc}	94.25	pu	m_p	2	%
p_i^*	1.0	pu	m_q	5	%
q^*	0	pu	ω_0	$60 \cdot 2\pi$	$\frac{rad}{s}$
$ Z_{vi} $	1.66	pu			

digital signal processor, (i.e., V_{vi}). The results clearly show that the system dynamics of VI-Lim and SatLim GFM inverters during and after the disturbances are comparable when they use the same φ_{vi} , thereby validating the proposed unified model and demonstrating the large-signal equivalence of the VI-Lim and SatLim approaches with the same limiter angle. Furthermore, as shown in Case 1 (see Fig. 20(a)–(d)), for $\varphi_{vi} = 0^\circ$, the system loses synchronism with the grid under the frequency drop, which is indicated by the negative power injections from the GFM inverters. In contrast, employing the more inductive limiter angle, $\varphi_{vi} = 50^\circ$, increases the large-signal stability margins (larger unstable equilibrium point) which enables the system to swiftly converge back to the nominal operating point without a loss of synchronism. Similarly, in Case 2 (see Fig. 20(e)–(h)), aside from the similar large-signal behavior between the VI-Lim and SatLim GFM inverters, more inductive limiters provide more reactive power support during the voltage disturbance. These findings further corroborate the theoretical analysis outlined in this work on the proposed unified model.

E. Large-Signal Analysis of Power Systems Using Multi-Inverter Network Simulations

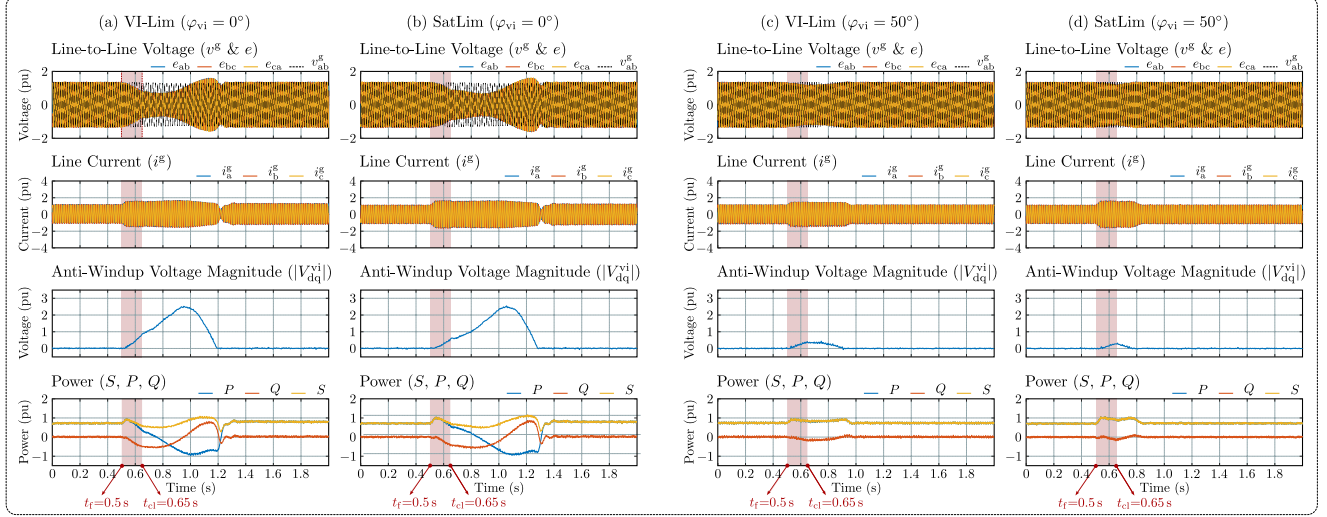
Leading up to this point, we analyzed large-signal stability for one GFM inverter connected to an infinite bus. In this section, we illustrate, through simulations, that general trends derived in the theoretical analysis of this paper carry over to network-wide fault conditions. We also illustrate and discuss the impact of the GFM inverter current-limiter settings on other network-wide attributes, including system stability and system-wide voltage support. The goal of this network-wide simulation effort is threefold. First, we aim to showcase the validity of the unified

current limiter model framework in a realistic network and grid disturbances. To that end, we consider the IEEE 14-bus network shown in Fig. 21, where five GFM inverters are interconnected, featuring 12 constant-impedance loads. The load values and line parameters are one-to-one translated from the standard IEEE 14-bus benchmark model. For a realistic mix of GFM inverter control topologies, we implement the dispatchable virtual oscillator primary controller (dVOC) in the GFM inverters at bus 1 and 4, the GFM inverters at bus 2 and 5 have the droop primary controllers, and the GFM inverter at bus 3 has the virtual synchronous machine (VSM) primary controller [34]. The active power setpoints for each of the GFM inverters are: $p_1^* = 0.5$ pu, $p_2^* = 0.2$ pu, $p_3^* = 0.9$ pu, $p_4^* = 0.4$ pu, and $p_5^* = 0.9$ pu. All inverters employ either the SatLim or VI-Lim current limiter, with a tunable limiter angle, φ_{lim} . In the middle of line 3, as illustrated in Fig. 21, we apply an unbalanced line-to-line fault at $t=0$ s that prolongs for 100 ms before being cleared, and we assess the system stability of the network during and after the fault. We deliberately apply an unbalanced fault as it presents a more realistic grid disturbance and allows for illustrating that the unified current-limiting framework remains valid even during unbalanced operating conditions. To be able to handle unbalanced conditions, we translate the GFM controls into the stationary reference frame, according to [9]. Second, we illustrate that the large-signal fault behavior of the GFM inverters is primarily dictated by the current-limiter angle, φ_{lim} ; therefore, in accordance with the proposed unified current-limiting modeling framework, SatLim and VI-Lim react similarly, if they employ the same limiter angle, φ_{lim} , even during network-wide disturbances. This clearly demonstrates its use for system-level transient-stability studies. Lastly, we illustrate that a more inductive (resistive) current-limiter angle increases (decreases) the stability margins during grid disturbances, independent of limiter types. We further illustrate that the current limiter—by manipulating the inverter output impedance during disturbances—has a direct effect on network-wide voltage support and system stability.

1) *System-Wide Transient Stability*: We consider two simulation cases: one, where all five GFM inverters employ a significantly resistive limiter, Lim¹⁰, and two, where all inverters leverage a primarily inductive limiter, Lim⁸⁵. Both cases are performed with the SatLim and VI-Lim (i.e., four simulations in total), allowing us to compare their respective dynamic response to the network fault. Inverter parameters are in Table III.

Fig. 22 portrays the EMT simulation results in the form of the GFM-inverter positive-sequence output voltage, E_x^+ , inverter active output power, P_x , inverter reactive output power, Q_x , internal reference frequency, $\omega - \omega_0$, and relative angle, $\theta - \theta_1$, of each of the five inverters, with θ_1 denoting the internal reference angle of GFM₁. Fig. 22(a) and (b) show the results when, respectively, a Lim¹⁰ and Lim⁸⁵ are leveraged in the GFM inverter controls. From Fig. 22, we observe: i) in accordance with the ideas postulated in the unified framework, the large-signal behavior of the GFM inverters when employing the SatLim or VI-Lim are comparable when their current-limiter angles are the same; ii) employing the predominantly resistive Lim¹⁰ (see Fig. 22(a)) causes the GFM inverters to lose synchronization with each other during and after the fault in the given fault

Case 1: Grid-side frequency jump from 60Hz to 59.4Hz for 0.15 seconds



Case 2: Grid-side voltage drop from 1.0 pu to 0.5 pu for 0.5 seconds

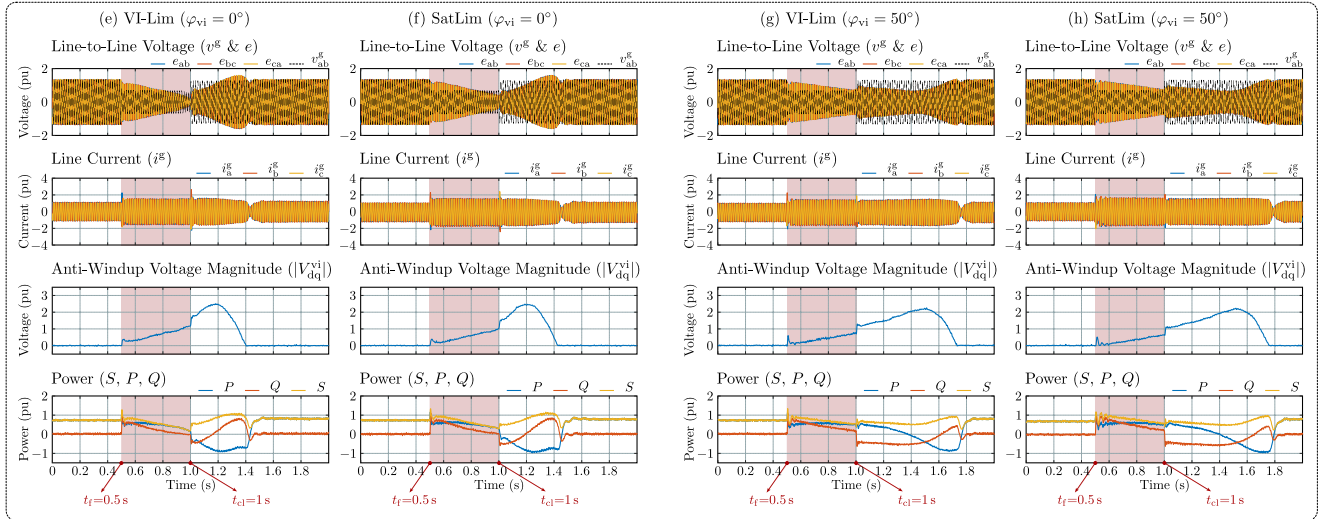


Fig. 20. Experimental results from the SMIB hardware tests under a frequency jump disturbance (Case 1) and grid-side voltage drop (Case 2).

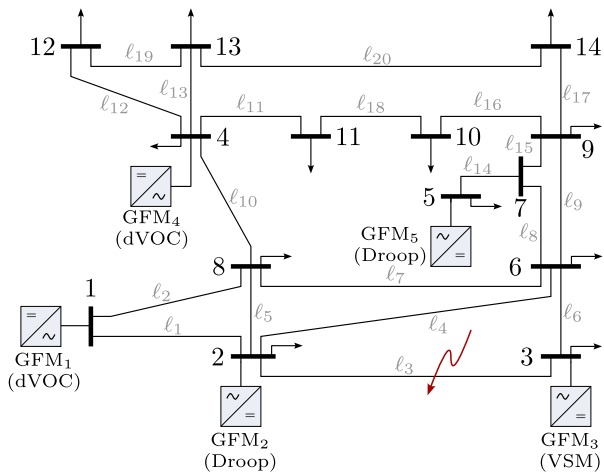


Fig. 21. IEEE 14-bus network with five GFM inverters interconnected while applying unbalanced line-to-line faults on line 3.

TABLE III
GFM INVERTER PARAMETERS FOR IEEE 14-BUS SYSTEMS

Parameter	Value	Unit	Parameter	Value	Unit
ω_0	$60 \cdot 2\pi$	$\frac{\text{rad}}{\text{s}}$	ω_i	$3000 \cdot 2\pi$	$\frac{\text{rad}}{\text{s}}$
$L_i \omega_s$	0.0196	pu	ω_v	$500 \cdot 2\pi$	$\frac{\text{rad}}{\text{s}}$
R_i	0.0139	pu	m_p (Droop)	2	%
$C \omega_s$	0.1086	pu	m_q (Droop, VSM)	5	%
$L_g \omega_s$	0.0196	pu	M (VSM)	0.5	s
R_g	0.0139	pu	κ_1 (dVOC)	0.0033	pu
$i_{\max}$	1.2	pu	κ_2 (dVOC)	0.0796	pu

ω_0 : nominal angular grid frequency; ω_i and ω_v : inner-current and outer-voltage control loop bandwidths; M : inertia constant in the VSM primary controller; κ_1 , κ_2 : synchronization gain and voltage-amplitude control gain in the dVOC primary controller [29].

condition. The bus voltages in the network collapse as a result. In contrast, employing the more inductive Lim⁸⁵ (see Fig. 22(b)) enables the various primary controllers of the GFM inverters to converge back to synchronization. Fig. 23 portrays the three-phase voltages and currents at the terminals of the GFM

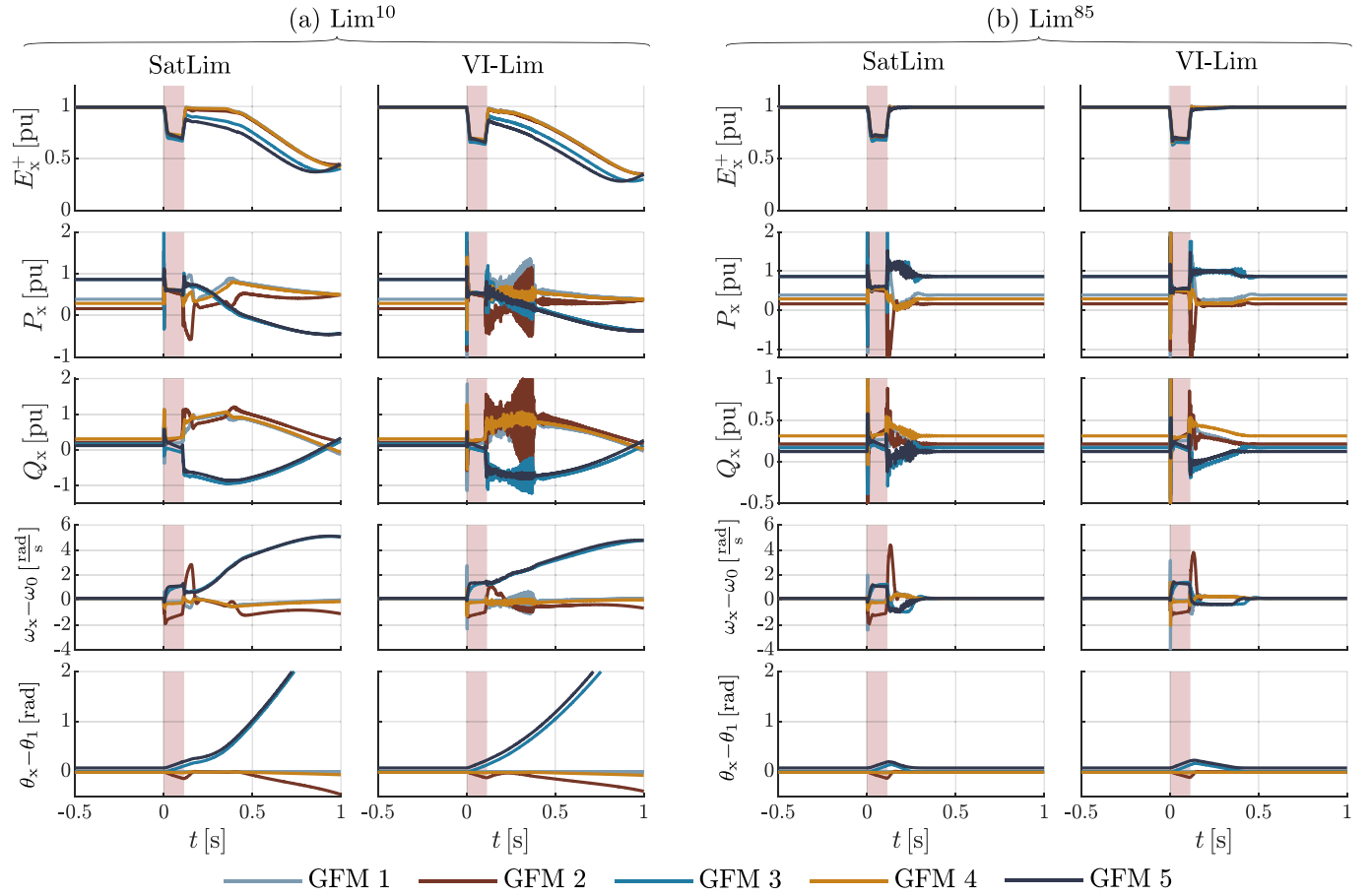


Fig. 22. IEEE 14-bus simulation results for an unbalanced fault at line 3: (a) results when employing a Lim^{10} , and (b) results when employing a Lim^{85} , implemented via the SatLim (left) and the VI-Lim (right).

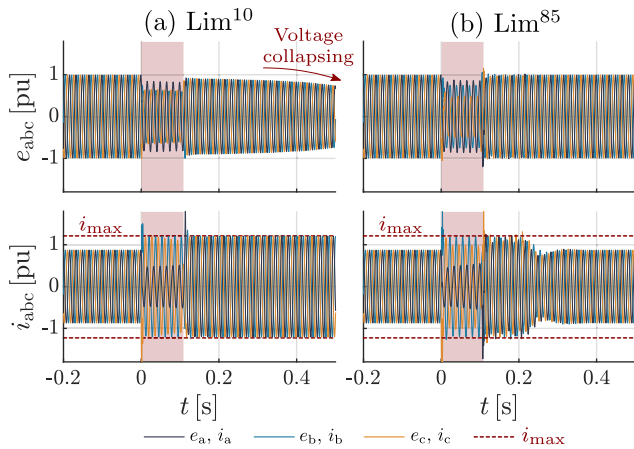


Fig. 23. Voltages and currents at bus 3 of the IEEE 14-bus simulation: (a) when employing the Lim^{10} and (b) Lim^{85} in the GFM inverters.

inverter connected to bus 3 for the SatLim case with (a) Lim^{10} and (b) Lim^{85} , and illustrates the unbalanced nature of the fault. Notice that the line currents are neatly limited to $i_{\max}$ by the current limiter. Further, the more inductive limiter, Lim^{85} , yields a fast and stable fault recovery, which agrees with the inverter-level analysis and validates its use for system-level studies.

2) *Network-Wide Voltage Support:* As explained in the introduction, the presence of a current limiter in a GFM inverter poses

various challenges affecting both inverter-level and system-level attributes. Apart from the effects on large-signal stability, GFM inverter current limiters and their design have a direct impact on, e.g., network-wide voltage support and compatibility with traditional power system protection systems. The unified current-limiter framework can serve as a tool to analyze these network-wide ripple effects. In this section, we illustrate this for grid voltage support. Once more, consider the IEEE 14-bus network in Fig. 21 with an unbalanced fault on line 3. To better illustrate the effect of the current-limiter angle on the network-wide voltage support, we decrease the grid strength by uniformly increasing all the line impedances by a factor of two, since the voltage degradation is more pronounced in weaker grids. We simulate the fault four times while changing the equivalent current-limiter angle from $\varphi_{\text{lim}} = 0^\circ$ to $\varphi_{\text{lim}} = 60^\circ$. Fig. 24 portrays the positive and negative sequence voltages at each of the inverter buses during the unbalanced fault (data was extracted after the initial transients have died out). The bar charts convey that, because of the inductive nature of the network, higher current-limiter angles result in better voltage support, i.e., higher (lower) positive (negative) sequence bus voltages. This is a direct consequence of the influence that the current limiter has on the output impedance of the inverter during the disturbance. Note that the objective of this simulation is not to optimize the current limiter design for network-wide voltage support, but to illustrate how the presented unified current-limiter framework can serve as a design and analysis tool.

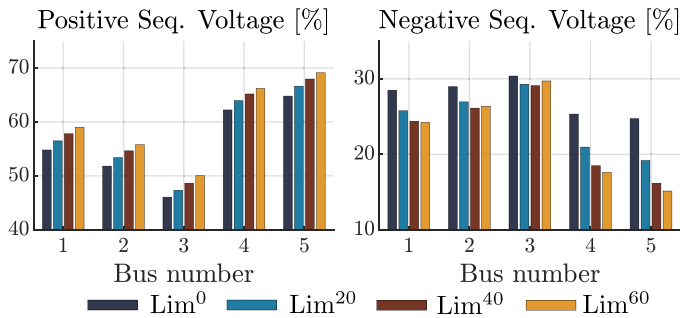


Fig. 24. Positive and negative sequence bus voltages for various current-limiter settings during the unbalanced fault in line 3.

These network-wide simulations of unbalanced grid disturbances demonstrate the use of the unified current-limiter modeling framework in system-level analysis. This framework can effectively enhance the understanding of critical system behaviors during transients, such as network stability, voltage support, and system protection.

VI. CONCLUSION

The current limiter in a GFM inverter is an essential part of the control system to protect the inverter against overcurrents caused by system disturbances. As a side effect of curtailing the output current during contingencies, the current limiter dynamically alters the output impedance of the GFM inverter. This has significant implications for network-wide attributes and systems, such as network-wide stability, inverter synchronization with the grid, power system protection, and dynamic network voltage support. To deepen the understanding of GFM inverter current limiting and its effects on large-signal stability, this paper presented a unified current-limiter model. The model incorporates two common current-limiting methods into one continuous algebraic model, which enables various theoretical analyses, such as large- and small-signal stability analysis and model order reduction. From the unified model and building on classical power system stability concepts, we derived an analytical expression for the power-angle characteristics of GFM inverters under various current-limiting conditions. This theoretical framework elucidates the mechanisms of large-signal stability in GFM inverters and the role of current limiting in exacerbating instability in a refreshing and succinct manner. We validated the proposed unified model and analytical model-based power-angle characteristics using numerical computation, full-order EMT simulations for various fault scenarios, and hardware experiments. The system-level stability and network-wide voltage support studies illustrated how the proposed unified current-limiter model framework can serve as a tool for system-level analyses. The multifaceted modeling and analysis confirm the validity of the unified models of current-limiting GFM inverters for device-level and system-level large-signal studies. Notably, using the unified model, this work revealed the equivalence of inner-control-based current-limiting methods. Regardless of the limiter type, e.g., SatLim or VI-Lim, their large-signal behavior can be characterized by their impedance angle. Future work includes extending the unified current-limiter model to more current-limiter types and exploring the applications of the model for large power system studies in greater detail.

VI. ACKNOWLEDGMENT

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